

VerdictTrace: Tamper-Evident, Byzantine-Robust Cross-Layer Drift Attribution for Connected Vehicle Fleets

1. Problem Statement

Design and evaluate an edge-deployed intrusion detection system for connected vehicle fleets that (a) detects concept drift independently on the in-vehicle CAN bus and the external V2X mesh, using cross-layer correlation as one input — not a standalone verdict — to help discriminate benign change from coordinated adversarial activity; (b) shares only per-vehicle attribution verdicts and confidence scores across the fleet, reaching Byzantine-robust consensus, with an explicit minimum-quorum requirement and defined fallback behavior when nearby vehicles are too few to vote meaningfully; and (c) detects tampering with a vehicle's own audit trail — distinguishing genuine tampering from ordinary connectivity gaps and ordinary model forgetting — while a delay-tolerant witnessing layer reduces, but does not claim to eliminate, the evidence gap left by disconnected and subsequently compromised vehicles.

2. The Problem, In Detail

2.1 Detection. CAN and V2X drift continuously from legitimate causes; both can be targeted by an attacker inducing slow, attack-like drift. **Correction made here:** the original design treated single-layer drift as automatically benign. This is wrong — the single most common real-world car-hacking pattern (an OBD-II port injection attack touching only CAN, never V2X) would have been waved through under that rule. Single-layer change is genuinely ambiguous, not safe, and correlation across layers should raise or lower confidence in a per-layer verdict, not replace the per-layer detector's own judgment.

2.2 Federation. Federating raw model weights is exploitable (Uddin, Nayeem & Bhuiyan, 2026, show this survives state-of-the-art robust aggregation at 30–40% malicious clients). Federating a scalar verdict instead shrinks the attack surface — but **only if enough vehicles are actually voting**. The original design never specified a minimum quorum; Byzantine-robust methods are statistically meaningless with two or three voters.

2.3 Evidence. An attacker who already succeeded has incentive to erase the trace — by editing logs, or, since this system uses continual learning, by feeding the model engineered updates to induce forgetting of a previously-caught pattern (established as a general ML attack by Umer et al. 2020 and PACOL 2023, not automotive-specific). **Correction made here:** the original log-tamper rule ("mismatch = tampering") ignored that mismatches also arise from ordinary connectivity loss, and the original model-memory rule stated a detection criterion more confidently than the underlying problem is actually solved.

3. System Design

3.1 Core

- **Testbed:** Raspberry Pi + OBD-II/CAN shield generating real CAN traffic, paired with a simulated V2X client, common clock.
- **Cross-layer drift-attribution engine (corrected):** per-layer detectors independently judge whether a change on their own layer looks benign or attack-like, using gradualness, match to known update signatures, and timing. Cross-layer correlation is then applied as a **confidence modifier**: correlated timing across both layers raises suspicion; no correlation leaves the per-layer verdict as the primary signal rather than defaulting to "safe."
- **Verdict-federation + Byzantine-robust consensus (corrected):** a defined minimum quorum of nearby voting vehicles is required before consensus is treated as reliable. Below quorum, the system falls back to local-only detection with explicitly lower confidence, rather than running robust aggregation over too few data points and reporting it as trustworthy.
- **GhostTrace, log layer (corrected):** three-way outcome instead of two. (1) Local and coordinator logs match → clean. (2) Mismatch explainable by a confirmed connectivity gap (no message was ever sent/received) → benign/ambiguous, not flagged. (3) Mismatch where a message was confirmed transmitted but its recorded content now differs → flagged as tampering. Only case 3 triggers an alert.
- **Synchronized CAN+V2X drift dataset**, cause-labeled, released as a standalone artifact.
- **Core evaluation:** chronological splits, Pi-4-class hardware, direct re-run of Bhuiyan et al.'s attack types against verdict-federation, SEED/ReCDA as continual-learning baselines.

3.2 Stretch

- **GhostTrace, model-memory layer (corrected framing):** periodic re-testing against previously-confirmed attack signatures to detect induced forgetting. **Now stated honestly as an open research question, not a solved detection rule** — ordinary catastrophic forgetting produces the same surface signature (degraded detection, no logged benign-drift event) with zero attacker involved, and distinguishing the two is the actual unsolved problem this component exists to address, not something the write-up should imply is already resolved.
- **V2X-based delay-tolerant forensic witnessing (corrected framing):** signed receipts broadcast to nearby vehicles, retained independently, propagated store-and-forward when reconnecting. Witness-based evidence and store-and-forward propagation are established techniques in vehicular delay-tolerant network security (Dias et al., 2015; encounter-ticket schemes, ~2010) — cited explicitly as prior art, not claimed as new. **Honesty correction:** the mechanism helps least exactly when it's needed most, since disconnection from the coordinator and isolation from nearby vehicles tend to co-occur (a rural dead zone has both no signal and few nearby cars). The write-up now states this as a **density-dependent reduction** of the evidence gap, not a solution — in near-zero-density scenarios, the system explicitly degrades to log-only detection with lower confidence rather than implying full coverage.
- **Full timeline reconstruction** (vs. detection-only) — hardest, least-precedented piece, optional polish.
- **Audit-trail tampering dataset**, labeled (deletion, timestamp rollback, selective falsification, induced forgetting).

4. Threat Model

(a) single-vehicle slow-poisoning of local attribution; (b) a compromised vehicle lying in the verdict vote; (c) coordinated Sybil voting, including below-quorum manipulation; (d) post-hoc tampering — log-level (with connectivity gaps correctly excluded) and, if stretch work lands, model-memory-level (with ordinary forgetting correctly excluded).

5. Metrics

Macro-F1, zero-day recall, FPR, forgetting score, drift-detection delay, drift-attribution accuracy, cross-layer correlation gain, attribution accuracy under adversarial lying, quorum-threshold sensitivity (how consensus reliability changes with vehicle density — new, follows directly from the quorum fix), and, if stretch work lands, tamper-detection accuracy with false-positive rate specifically on connectivity-gap and ordinary-forgetting cases (this is the metric that proves the corrected three-way logic actually works, not just the headline detection number).

6. Research Gaps — Verification Status, Fully Honest

Claim	Status
Cross-layer CAN+V2X drift	Checked — no equivalent found
Verdict-federation vs. weight-federation	Checked against directly-named 2026 competitor
Continual-learning IDS audit-trail	Checked against closest paper (general sensor tampering, not IDS verdict logs)
Model-memory tampering as	Mechanism established (2020/2023), automotive application not — now correctly framed as open, not solved
V2X witnessing for ML-IDS evidence	Mechanism established (2010–2015), application to ML attribution verdicts not — now correctly framed as density-limited
Not independently verified	VANET-crypto (<i>Scientific Reports</i> , 2024), DOT report, IEEE TITS 2025 V2X security survey — read the survey's related-work section before

7. Team Split, Phased

Core (months 1–11): A — CAN drift detection (corrected three-case logic); B — V2X sim + testbed; C — verdict-federation + quorum-aware Byzantine consensus + GhostTrace log layer (three-way logic); D — dataset pipeline + evaluation harness. Stretch (months 12–16, contingent on core done and tested): model-memory GhostTrace or witnessing, not both unless core finishes early. Months 17–18: writing, buffer.

8. References — Full List

Uddin, Nayeem & Bhuiyan, "Robust Federated Learning for Anomaly Detection in Connected Autonomous Vehicle Networks Under Adversarial Attacks," *Automation*, 2026, 7(3), 80. · Yang et al., "SEED: Semi-supervised Continual Malware Detection for Tackling Concept Drift on a Budget," 2025. · Yang et al., "ReCDA: Concept Drift Adaptation with Representation Enhancement for Network Intrusion Detection," ACM SIGKDD, 2024; related IEEE TDSC 2025 self-supervised drift-adaptation paper. · Korycki & Krawczyk, "Adversarial Concept Drift Detection Under Poisoning Attacks," 2020. · "In-vehicle device data tampering detection: Accurate identification based on correlation calculation and data relationship," *ScienceDirect*, Aug 2025. · Block4Forensic, "An Integrated Lightweight Blockchain Framework for Forensics Applications of Connected Vehicles," 2018. · Umer, Dawson & Polikar, "Targeted Forgetting and False Memory Formation in Continual Learners through Adversarial Backdoor Attacks," 2020 (arXiv:2002.07111). · "PACOL: Poisoning Attacks Against Continual Learners," 2023 (arXiv:2311.10919). · Dias, Rodrigues, Xia & Mavromoustakis, "A Cooperative Watchdog System to Detect Misbehavior Nodes in Vehicular Delay-Tolerant Networks," *IEEE Trans. Industrial Electronics*, 62(12), 2015. · **Unverified, confirm before submission:** V2X-VANET cryptographic-protocols paper (*Scientific Reports*, 2024); U.S. DOT connected-vehicle cyber resilience report; "Advancing Intrusion Detection in V2X Networks" survey (*IEEE Trans. Intelligent Transportation Systems*, 2025).